The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Supplementary Material

Appendices and supplementary figures accompanying the article. Section, equation and figure numbers referenced here follow the main text; supplementary figures are numbered S1, S2, ...

A Notation and units

These neural-mass models are phenomenological: potentials are in mV, firing rates in s^{-1} (≡Hz), time constants in ms, and frequencies in Hz, unless noted; a "–" marks a dimensionless quantity. A few glyphs are reused across the four models (JR, LaNMM, NMM2/MPR, QIF) by long-standing convention; these are disambiguated by context and flagged below. In particular, A without an argument is the excitatory PSP *gain* in the JR/LaNMM synapses; $A(t)$ is the instantaneous envelope of the two-tone field and A_Ω a lock-in amplitude. The applied-field modulation amplitude is ε in JR and LaNMM alike, and A_f in NMM2/QIF. These are not interchangeable units. In JR and LaNMM the field is a membrane polarization in mV entering the sigmoid argument, so ε is measured against the sigmoid voltage scale $1/\rho \approx 1.8$ mV, which is the same in both models. In NMM2/QIF the field enters the excitatory membrane current in the dimensionless MPR units of v_E, and A_f carries no volt scale at all. Likewise C is the JR connectivity constant and the NMM2 global synaptic gain $C\,(\equiv J)$.

Symbol	**Meaning**	**Units**
Firing-rate nonlinearity (sigmoid)		
$\sigma(v)$	population firing-rate transfer function	s^{-1}
$\sigma'(v)$	sigmoid slope (read-out gain)	$s^{-1}\,mV^{-1}$
$\sigma''(v)$	sigmoid curvature; the memoryless detector coefficient	$s^{-1}\,mV^{-2}$
$F(Q)$, F', F''	MPR steady-state rate curve, Eq. (5), and its derivatives	–
$C(Q,\omega)$, $K(\omega)$	MPR second-order DC coefficient at fixed current, Eq. (12), and at fixed voltage excursion, Eq. (14); d, ν the population's relaxation rate and ringing frequency	–
e_0	half the maximal firing rate ($\sigma \to 2e_0$ as $v \to \infty$)	s^{-1}
v_0	sigmoid inflection (half-maximal) potential	mV
ρ	sigmoid slope parameter	mV^{-1}
v^*	operating point (field-free fixed point of v)	mV
$v = y_1 - y_2$	net pyramidal membrane potential (LFP proxy)	mV
PEIX	Population Excitation–Inhibition Index (a σ'-based brain-state index)	–
Single-column Jansen–Rit		
A, B	excitatory, inhibitory PSP gains	mV
a, b	excitatory, inhibitory synaptic rates (inverse time constants)	s^{-1}
y_1, y_2	excitatory, inhibitory post-synaptic potentials	mV
C, $C_1 - C_4$	intracolumn connectivity constants	–
p	external input rate; JR bifurcation parameter	s^{-1}
Applied TI field and coupling		
ε	field modulation amplitude, JR and LaNMM (a membrane polarization entering the sigmoid argument)	mV
A_f, $I_E(t)$	amplitude of the field's current-like forcing $I_E(t) = P\,E(t)$ in the E voltage equation, NMM2/QIF (dimensionless MPR units of v_E)	–
m	modulation depth	–
f_c, $\omega_c = 2\pi f_c$	carrier frequency	Hz, $rad\,s^{-1}$
f_1, f_2	the two applied carrier tones ($f_c = (f_1 + f_2)/2$)	Hz
Δf, $\Omega = 2\pi\Delta f$	envelope (beat) frequency, $\Delta f = f_2 - f_1$	Hz, $rad\,s^{-1}$
$E(t)$, $s(t)$	applied field; membrane perturbation reaching the nonlinearity ($s = \sum_c \boldsymbol{\lambda}_f^c \cdot \mathbf{E}$, Eq. (7))	$V\,m^{-1}$; mV
λ, λ_0^c, λ_f^c	field-to-membrane coupling: generic, DC (door c), and carrier-effective ($\lambda_f^c = \lambda_0^c \lvert H_c \rvert$)	$mV\,/\,(V\,m^{-1})$
$\Lambda[\hat{E}]$	direct (non-synaptic) field term in the sigmoid argument, Eq. (3); equals $s(t)$ in the JR column	mV
$H_c(\omega)$, H_{gate}	access filter of door c; Na_v activation-gate filter	–
$H_s(\omega)$	synaptic (second-order) transfer function, §2.5	–

Symbol	Meaning	Units
$H_m(\omega),\ \tau_m$	somatic membrane low-pass and its time constant	–; ms
τ	fast-element (node/AIS) time constant; also $\tau = 1/\gamma$ in $\mathcal{Q} = \pi f_0 \tau$	ms
Network dynamics, response, and bifurcation		
$\lambda_\pm = -\gamma \pm i\omega_0$	leading Jacobian eigenvalue pair	s^{-1}
γ	damping of the critical mode ($\propto$ distance to the Hopf)	s^{-1}
$\omega_0,\ f_0 = \omega_0/2\pi$	natural (Hopf) frequency	$\mathrm{rad\,s}^{-1}$, Hz
$\mathcal{Q} = \omega_0/2\gamma = \pi f_0 \tau$	resonance quality factor ($\tau = 1/\gamma$ the reverberation time)	–
$G(\Omega)$	network resonance gain (Lorentzian, $\propto 1/\gamma$ at ω_0)	–
A_Ω	demodulated response: lock-in amplitude at Ω of v (mV) or, in §4.7 and the QIF network, of a rate	mV or s^{-1}
Y_Ω, Y_{LF}	detector (pre-network) output at Ω and at baseband, §2.4	–
$\mathcal{M}[\cdot]$	peak-to-peak excursion of a periodic quantity, Eq. (17)	–
η_{det}	TI/tACS drive ratio of a memoryless transfer at matched peak polarization, App. D	–
δ	threshold offset of a step detector, App. D	mV
$\ell(\omega)$, ℓ_0	AC space constant and its DC value, Eq. (22)	μm
PPC	pairwise phase consistency of single-neuron spike phases at Δf [1]	–
A_{open}	same lock-in, open loop (detector only, no recurrence); $A_\Omega/A_{\mathrm{open}}$ dimensionless	mV
I, Q	in-phase, quadrature lock-in components	mV
C^*, J^*	coupling at the Hopf (criticality)	–
Laminar model (LaNMM)		
P_1, P_2	deep (alpha) and superficial (gamma) pyramidal populations	–
PV, SS, SST	parvalbumin, stellate, somatostatin interneurons	–
Exact mean field (NMM2 / MPR) and QIF network		
r_E, r_I	population firing rates (E, I)	s^{-1}
v_E, v_I	mean membrane potentials (E, I), in dimensionless MPR units	–
$\eta_j,\ \bar{\eta}$	per-neuron excitability; mean excitability (= common drive, bifurcation param.)	–
Δ	Lorentzian half-width of η_j (excitability heterogeneity)	–
$\tau_E, \tau_I, \tau_a, \tau_g$	E, I membrane and AMPA, GABA synaptic time constants	ms
$C\ (\equiv J)$	global synaptic coupling (inter-population gain)	–
$A_{ee}, A_{ei}, A_{ie}, A_{ii}$	synaptic weight matrix $(1, 1, 1, 2)$	–
s_E, s_I	second-order synaptic activations; $\hat{K}_a, \hat{K}_g$ their operators	s^{-1}
$V_j,\ V_{\mathrm{peak}}$	QIF neuron potential; spike/reset threshold	mV
N	neurons per population (QIF; $N = 8000$)	–
$A_\Omega/\langle r_E \rangle$	Δf modulation depth (envelope-locked timing metric)	–

Table 1: Symbols, meanings, and (nominal) units. Reused glyphs are disambiguated by context; see the head note on A and C.

B Simulated models: equations and parameters

The models are written in the NMM1 transfer-function formalism of Eq. (3) [2]: synaptic operators $\hat{L}$ transduce firing rates into PSPs, the PSPs sum with any direct field term $\Lambda[\hat{E}]$ in the sigmoid argument, and the sigmoid returns the rate. The Jansen–Rit column is given here; the laminar model is specified in [3, 4] with its field drive in Eq. (25), and the NMM2 PING motif and its QIF network in Eq. (26) and §3.4.

Jansen–Rit column

Writing y_0 for the excitatory PSP driving the interneurons, y_1 and y_2 for the excitatory and inhibitory PSPs onto the pyramidal population, external input p, and the applied-field perturbation $s(t)$ (§2.2) entering the pyramidal sigmoid,

$$\ddot{y}_0 = Aa\,\sigma\big(y_1 - y_2 + s(t)\big) - 2a\dot{y}_0 - a^2 y_0, \tag{1}$$

$$\ddot{y}_1 = Aa\,[\,p + C_2\,\sigma(C_1 y_0)\,] - 2a\dot{y}_1 - a^2 y_1, \tag{2}$$

$$\ddot{y}_2 = Bb\,C_4\,\sigma(C_3 y_0) - 2b\dot{y}_2 - b^2 y_2, \tag{3}$$

integrated as six first-order ODEs; the LFP proxy is $v = y_1 - y_2$ [5].

Table 2: Jansen–Rit parameters [5].

Symbol	Meaning	Value
A, B	excit./inhib. PSP gain	3.25, 22 mV
a, b	synaptic rates	100, 50 s^{-1}
v_0, e_0, ρ	sigmoid inflection, max rate/2, slope	6 mV, 2.5 s^{-1}, 0.56 mV^{-1}
C	global connectivity	135
C_1, C_2, C_3, C_4	couplings	C, $0.8C$, $0.25C$, $0.25C$
p	external input (bifurcation knob)	0–400 Hz

C TI requires detection: the Stuart–Landau normal form amplifies but cannot demodulate

Amplification and detection are separable, and the universal Hopf normal form makes the separation sharp. The Stuart–Landau equation [6] is

$$\dot{z} = (\alpha + i\omega_0)z - |z|^2 z + \mathcal{F}(t), \qquad \gamma \equiv -\alpha > 0 \ \ \text{(stable side)}, \tag{4}$$

driven by the TI field $\mathcal{F}$. Expanding $z = \sum_k Z_k(t)e^{ik\omega_c t}$ in carrier harmonics with slowly varying envelopes, the field forces only $k = \pm 1$ and the cubic feeds index $k = l - p + q$. The odd-k subspace is therefore invariant and attracting: if every even Z_k vanishes, then for even k both sources vanish—$\mathcal{F}_k = 0$, and $l - p + q$ cannot be even with l, p, q odd—leaving $\dot{Z}_k = [\alpha + i(\omega_0 - k\omega_c)]Z_k$, which decays at rate $\gamma > 0$. So z lives only on odd carrier harmonics, to all orders in ε. The beat line is a *baseband* signal and would sit in $k = 0$; since $Z_0 \equiv 0$, the response at Δf is identically zero. An *even* nonlinearity instead produces a nonzero $k = 0$ term through $\cos^2 \omega_c t \to \frac{1}{2}$—the very line the odd cubic cannot make. Numerically ($\alpha = -0.5$, $f_0 = 10$, $f_c = 100$ Hz, $\varepsilon = 0.5$), a square-law detector on the same field gives a Δf lock-in of 2.5×10^{-1} while $\mathrm{Re}\, z$ gives $\sim 2 \times 10^{-8}$, at the numerical floor. Driven instead directly in band, the same equation shows the forced $1/\gamma$ growth toward the Hopf.

The normal form thus responds to tACS but is blind to TI. Amplification is the model-independent near-Hopf susceptibility, shared with any normal-form resonator; detection is the model-specific even-order term the normal form lacks. A critical resonator alone cannot demodulate.

D Efficiency of a memoryless transfer against a direct in-band drive

This appendix keeps the bound on the TI/tACS ratio for a bounded monotone *memoryless* transfer, at matched peak post-coupling polarization and balanced tones. It is a statement about that comparator: it does not govern the dynamic detector of §2.3, whose source is a voltage-domain mixing term of a different dimension, and it is not a limit on TI in general. Writing the TI drive $\frac{1}{2}\sigma''\varepsilon_1\varepsilon_2$ against the tACS drive $\sigma'\varepsilon$ at the same peak polarization, with $\varepsilon_1 = \varepsilon_2 = \varepsilon$, the ratio is $\eta_{\mathrm{det}} = \sigma''\varepsilon/4\sigma'$, proportional to ε: a square-law detector gets less efficient as the signal weakens.

The obvious escape is the operating point, since both σ'' and σ' depend on it and the column might sit somewhere far more favorable. It does not escape. For the logistic transfer, writing $s = \sigma/2e_0 \in (0,1)$, we have $\sigma' = 2e_0\rho\, s(1-s)$ and $\sigma'' = 2e_0\rho^2 s(1-s)(1-2s)$, so $\sigma''/\sigma' = \rho(1-2s)$ and, since $\mathrm{PEIX} = 4s(1-s)$ gives $(1-2s)^2 = 1-\mathrm{PEIX}$,

$$\eta_{\mathrm{det}} = \frac{\rho\,\varepsilon}{4}\sqrt{1-\mathrm{PEIX}} \;\le\; \frac{\rho\,\varepsilon}{4}. \tag{5}$$

The efficiency is therefore state-dependent exactly as the mechanism predicts—maximal where the population is saturated and far from the inflection, and vanishing at $\mathrm{PEIX} = 1$, where the transfer is locally linear and cannot demodulate at all—but at fixed ρ it is *bounded* by $\rho\varepsilon/4$ whatever the state. The maximum is one of ratio rather than of absolute response: deep in the saturated tails σ' and σ'' both vanish and the TI and tACS drives disappear together. At the physiological Jansen–Rit slope ($\rho = 0.56$ mV^{-1}, transfer width $1/\rho \approx 1.8$ mV) that ceiling is 1.4% at Vieira et al.'s fields and 0.28% at human ones. Brain state moves the efficiency, but at fixed ρ it cannot lift the transfer off that ceiling.

The slope is itself a parameter, so the question is what the whole sigmoid family can do. The logistic transfer depends on voltage only through $\rho(v - v_0)$, so writing

$$u = \rho\,\varepsilon, \qquad x = \rho\,(v^* - v_0), \tag{6}$$

we have $\sigma(v^* + \varepsilon q(t)) = 2e_0\, L(x + u\, q(t))$ for any normalized waveform q, with $L(z) = (1+e^{-z})^{-1}$. Every Fourier amplitude, and hence their ratio, is a function of (u, x) alone; the factor $2e_0$ cancels. Maximizing over v^* and ρ at fixed ε is therefore the same as maximizing over (u, x), and the family supremum is independent of field strength. Evaluating the exact cycle response rather than its quadratic truncation and optimizing x at each u, the ratio rises monotonically over the range explored and converges to the hard-threshold limit

$$\sup_{\rho,\, v^*} \eta_{\mathrm{det}} = 0.3479, \tag{7}$$

approached as $\rho \to \infty$ at a threshold offset δ (the distance from the operating point to the inflection) with $\delta/2\varepsilon = 0.523$. There is no finite optimal transfer width: narrower is always better, and reaching 95% of the ceiling requires $u \gtrsim 6$, that is $1/\rho \lesssim \varepsilon/6$.

Two features of that limit decide the reading. Such a detector is not weak. At the optimum, and at the Vieira-scale polarization $\varepsilon \simeq 0.1$ mV, its open-loop beat drive exceeds the physiological-slope sigmoid's by some three orders of magnitude for the same e_0 and waveform, because the field excursion is about twice the threshold offset and the transfer switches across its full range instead of responding linearly. The disparity grows as the field weakens, the smooth sigmoid's response falling as ε^2 while a threshold repositioned with the field keeps its excursion. The obstacle is the width. At Vieira et al.'s fields the ceiling is approached only for $1/\rho \lesssim 17$ μV, and at human polarizations for $1/\rho \lesssim 3$ μV, with the threshold offset held to the same order as the applied polarization. The width of a neural-mass sigmoid is set by the spread of thresholds and noise across $\sim 10^4$ neurons and is of order millivolts. A microvolt width lies far below that, and reads more naturally as an idealized microscopic threshold or rectifier than as a neural-mass sigmoid; we do not offer it as a measured single-cell property, real thresholds being themselves noisy and

state-dependent. The regime in which the family approaches its ceiling is thus one in which the "population sigmoid" has ceased to describe a population.

The limit is a step detector, not an envelope follower. For a unit step at offset δ the carrier-averaged transduction is $\pi^{-1}\arccos(\delta/A)$ for $A > \delta$ and zero below; with $\beta = \delta/2\varepsilon$ its tACS fundamental is $(2/\pi)\sqrt{1-\beta^2}$ and its TI coefficient reduces by parts to $\eta_{\text{det}} = 2\beta\,[K(m) - E(m)]/(\pi\sqrt{1-\beta^2})$ with $m = 1-\beta^2$ and K, E the complete elliptic integrals, giving 0.3479 at $\beta = 0.523$ by direct quadrature and by the elliptic form alike (`threshold_bound_check.py`). At that optimum the step's second-to-first harmonic magnitude ratio is 0.151, not the 1/5 of a linear envelope follower. A bounded nondecreasing transfer is a positive mixture of shifted steps and both Fourier coefficients are linear in the transfer, so the step optimum bounds the family; the maximum itself is a numerical evaluation, not a certified bound.

E Extension: the laminar column at low carriers

This extension drives the two-band laminar column (LaNMM, §2.1) with carriers of 25–80 Hz and polarizations up to 3 mV, outside the perturbative regime of the main text and below the carriers TI uses. It is included for what it shows about a band-limited intermediary between field and slow generator; it does not carry the kHz argument.

The single JR column isolates and lets us analyze the mechanism; the laminar model (LaNMM, §2.1) confirms it in a cortical column with its *own* two resonances. Driving one population with the two-tone field of Eq. (25) and reading the alpha-band power of the deep (P_1, alpha) and superficial (P_2, gamma) pyramidal potentials (Methods, Fig. S2), we recover the same envelope resonance, now carrying the column's intrinsic rhythms.

As a consistency check, driving the deep P_1 (alpha) loop directly reproduces the alpha-band beat resonance of the JR column on both the stable-focus and limit-cycle sides, with the sweep and its code in the repository; what the laminar model adds is the carrier-dependent structure below.

One qualification governs the laminar sweeps. They run to $\varepsilon = 3$ mV on P_2, above the sigmoid voltage scale $1/\rho \approx 1.8$ mV, so they sit deliberately in a nonlinear, non-perturbative regime. They are a demonstration of tongues and frequency capture, not a verification of the weak-field square law; that verification is the low-amplitude JR work (Fig. 6b), which fits the square law at 0.05–0.2 mV, one to two orders of magnitude below $1/\rho$.

Sweeping the field amplitude ε and beat frequency Δf at a fixed carrier produces *nonlinear Arnold tongues*: alpha power in P_1 is maximal when Δf matches the deep alpha ($\sim$10 Hz) and grows with ε (Figs. S1a,b and S7a,b), envelope resonance and phase locking of the slow generator to the recovered beat component, as in the JR resonance curves but now in a two-band column. Outside the tongue the drive *reduces* alpha-band power: with the field on P_1, 95% of the amplitude–beat grid and 97% of the carrier–beat grid lie below the field-free baseline (Fig. S7), the drive pulling the autonomous alpha out of the 8–12 Hz band or suppressing it; with the field on P_2 the fractions are 44% and 31% (Fig. S1). The earlier version of these maps showed only the positive part of the change.

The carrier dependence illustrates the two coupling limits of §2.2–§2.6 within this model's 25–80 Hz carrier range. Sweeping carrier f_c against beat Δf: when the field is delivered to the *superficial* loop (P_2), strong P_1 demodulation requires a *double* resonance—f_c near the P_2 gamma ($\approx$44 Hz) *and* Δf near the P_1 alpha ($\approx$10 Hz) (Fig. S1c,d): the fast loop acts as a tuned radio-frequency (RF) front end that must itself be excited before it can pass a demodulated envelope to P_1, and off that gamma resonance the response collapses to $\sim$7% of its peak. When the field is delivered *directly* to the slow generator (P_1), the picture inverts: the envelope-locked response persists at every carrier from 25 to 80 Hz, falling only to $\sim$25% of its peak, and the ridge sits at $\Delta f \approx 10.5$ Hz even for a 60 Hz carrier (Fig. S7c)—broadband in f_c, as Eq. (9) predicts for direct coupling. This contrast requires the phase-sensitive read-out: because P_1 free-runs at its own alpha, band power alone buries a weak envelope-locked drive under an autonomous rhythm that is not phase-locked to the beat, and only the lock-in at Δf separates the two (Methods). TI's kHz carriers lie outside this model's range; what the contrast shows is that a carrier matched to a fast (gamma) network resonance couples best, while the envelope response persists, weaker, off resonance. This both confirms the JR mechanism and refines it: the "inert carrier" of §2.5 is the direct-coupling idealization, while a band-limited intermediary makes the carrier choice matter, consistent with frequency-tagging/intermodulation paradigms [7].

Concretely the LaNMM predicts that TI efficacy should (i) peak when Δf matches a region's intrinsic alpha; (ii) widen and strengthen with modulation amplitude (the tongue); and (iii) for fields delivered via superficial fast circuits, be largest when the carrier is matched to a gamma network resonance—e.g. two nearby gamma carriers summing to a $\sim$40 Hz-centered AM field—

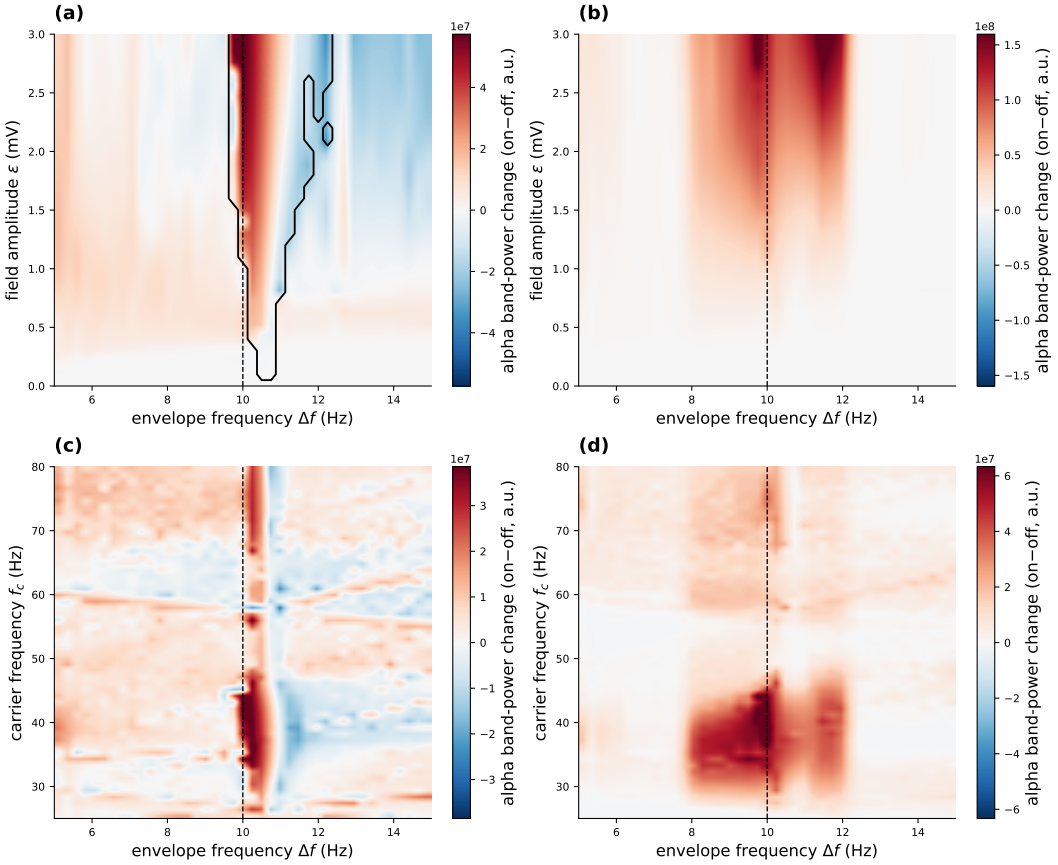


Figure S1: LaNMM Arnold tongues: two-tone TI field on the superficial P_2 (gamma) loop. The field enters the P_2 sigmoid argument as a membrane polarization ε (mV) with all external drives flat. Color shows the signed change of alpha-band (8–12 Hz) power from the unstimulated $\varepsilon = 0$ baseline, on a zero-centered scale (arbitrary units). *(a,b)* P_1 and P_2 enhancement vs. ε and Δf (carrier 40 Hz), peaking at the P_1 alpha (10.0 Hz, dashed). The black contour in *(a)* is the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix F), drawn because band power alone cannot separate a driven from an autonomous rhythm; it is a dominance screen, not a phase-capture test. The $\varepsilon = 0$ row is excluded, since with no drive the autonomous 10.67 Hz alpha already lies within the tolerance of two grid points (a resolution floor of 0.5 Hz); the set widens to 2.75 Hz at 3 mV. *(c,d)* Enhancement vs. carrier and Δf at $\varepsilon = 1.5$ mV: P_1 demodulation needs *both* a carrier near the P_2 gamma (≈44 Hz) and Δf near the P_1 alpha—a double resonance.

while remaining present, weaker, for off-resonance carriers. The third is untested: existing TI studies of deep targets sweep the *beat* frequency rather than the carrier against a network resonance [8, 9], so they neither support nor contradict it.

F Vocabulary and metrics

The three terms (forced response, entrainment, gating) are defined in Methods; this appendix gives the metrics behind them. *(i) Forced response.* Below the Hopf the column does not oscillate on its own; the field elicits a driven response whose susceptibility we quantify by the lock-in amplitude of the read-out at the beat Δf (Eq. (24)), and there is no autonomous rhythm to entrain (J-curve, timing, tACS). *(ii) Frequency entrainment.* Above the Hopf the column free-runs at f_0; genuine entrainment means its dominant frequency is *pulled* to Δf. We screen for it with the dominant output frequency f_{out} (largest non-DC FFT peak) and the criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz, mapping the tongue and locking range (Fig. S4); f_{out} is used rather than a phase-locking value [10] because, when an autonomous rhythm and a driven component coexist, instantaneous-phase estimates conflate them. The dominance criterion is a screen, not a capture test: the largest peak can move to the drive frequency because a driven component outgrows a persisting autonomous one. A capture test would follow the unwrapped phase difference and require the absence of secular drift; the maps therefore label the set the frequency-dominance set. *(iii) Envelope locking / gating.* A periodic drive can instead modulate the *amplitude* of a persisting rhythm at the beat without capturing its frequency—phase–amplitude coupling [11, 12]. We quantify this by the Δf-locked component of the population rate (lock-in of r_E at Δf, equivalently the rate *folded* on the Δf beat phase—averaged over many beat cycles as a function of phase within a single cycle, so that everything not locked to Δf averages away), which isolates the driven envelope from the intrinsic rhythm (the above-Hopf QIF panel, with its $\sim$11 Hz detuning, is a case of gating, not entrainment).

The Arnold-tongue protocol for Fig. S4 places the column on the limit-cycle side (input p between the SNIC and the Hopf, free-running at $f_0(p)$) and drives it with $s(t) = \varepsilon \cos \Omega t$; the implementation, including the grid widths and the clipping check, is in the released code. One subtlety: as onset is approached the cycle amplitude vanishes and the tongue merges into the forced-resonance bandwidth, so the supercritical locking range and the subcritical $1/\gamma$ susceptibility coincide there by construction—which is why responsiveness peaks at criticality on both sides.

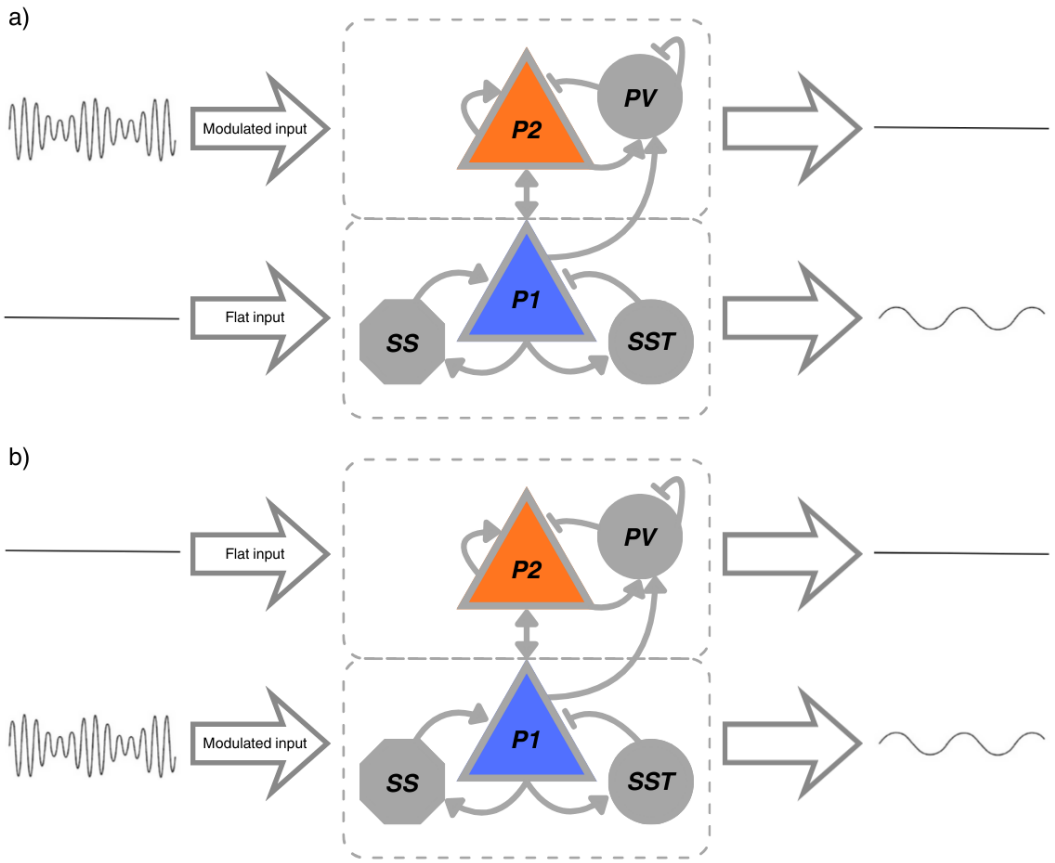


Figure S2: Stimulation setup in a LaNMM column. The column couples a superficial PING subcircuit (pyramidal P_2 + PV; gamma) and a deep JR subcircuit (pyramidal P_1 + SS + SST; alpha). *(a)* the TI field drives P_2 while P_1 receives a flat drive; *(b)* the TI field drives P_1 while P_2 is flat. Through the sigmoid nonlinearities and P_1–P_2 coupling, the generated beat-frequency component is transferred to the deep loop: P_2 carries the modulated fast carrier, while P_1 exposes the recovered low-frequency component (right traces).

G Supplementary figures

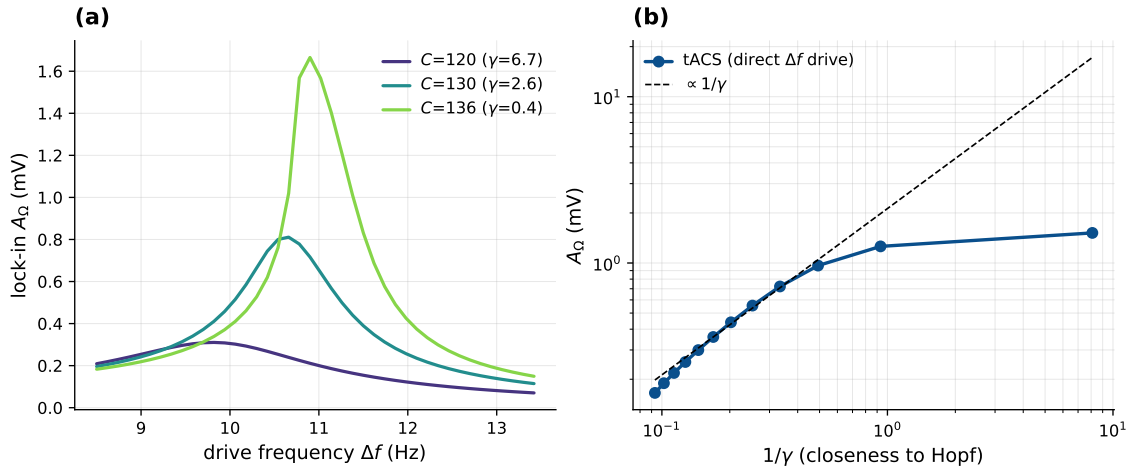


Figure S3: tACS shares the network amplifier (Jansen–Rit). A direct low-frequency (tACS) drive at Δf enters the column without a carrier, so it needs no demodulation, but feeds the same near-Hopf resonator. With the operating point pinned, sweeping the coupling $C\,(=J)$ toward the alpha Hopf: *(a)* the lock-in resonance at Δf sharpens and grows as $C \to C^*$; *(b)* the response grows as $1/\gamma$ (dashed) and saturates near criticality—the same network amplification as the demodulated TI drive (Fig. 9), but linear in the field rather than square-law. Throughout, the column is a stable focus ($\gamma > 0$).

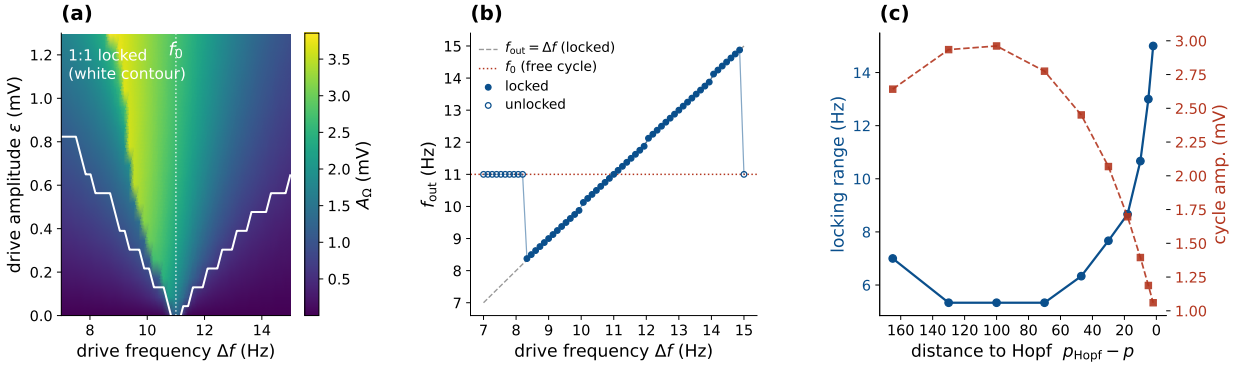


Figure S4: Above the Hopf: entrainment, not $1/\gamma$ amplification (Jansen–Rit). The column is set in its autonomous alpha limit cycle ($p = 250$, $f_0 \approx 11$ Hz) and driven by a direct in-band sinusoid. *(a)* The 1:1 Arnold tongue—where the dominant output frequency locks to Δf—is a wedge anchored at f_0 that widens with drive. *(b)* Inside the tongue the output follows Δf; outside, it falls back to f_0. *(c)* Toward the Hopf at fixed drive the locking range grows as the autonomous cycle amplitude vanishes (blue, left axis; red dashed, right axis). Above the Hopf amplification reads as tongue width rather than as the diverging $1/\gamma$ gain of the stable-focus side (Fig. 9): the column is most responsive near criticality on both sides.

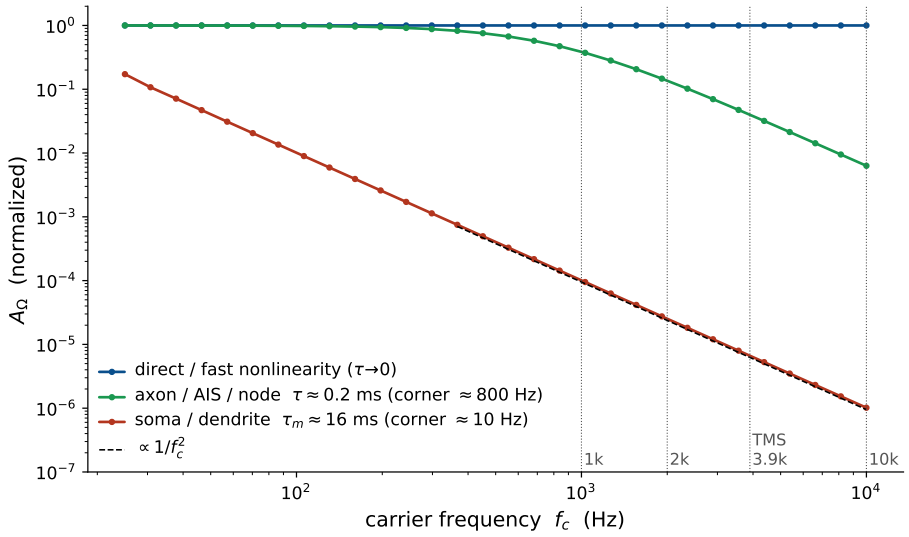


Figure S5: Somatic demodulation dies at kHz, but a fast element survives (Jansen–Rit, open-loop detector). Open-loop demodulated response (normalized to the $\tau \to 0$ value) vs. carrier frequency up to 10 kHz. With direct coupling ($\tau \to 0$) the response is flat—carrier-independent. A slow somatic membrane ($\tau_m \approx 16$ ms) rolls off as $1/f_c^2$ (dashed guide), killing kHz demodulation; a fast element (axon/AIS/node, $\tau \approx 0.2$ ms) retains an appreciable response into the kHz band ($\sim$0.14 at 2 kHz, $\sim$0.04 at the TMS peak 3.9 kHz). The same membrane filter attenuates TMS at the soma, which is why neither TMS nor TI should be thought of as driving the soma at kHz.

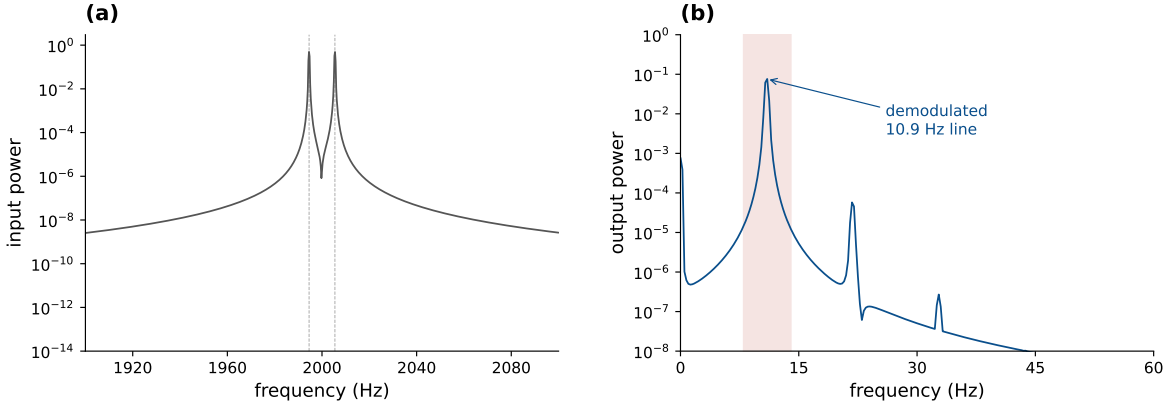


Figure S6: Genuine two-tone TI at a real kHz carrier (Jansen–Rit, fast-element route). Two equal tones at $f_1 = 1994.54$ and $f_2 = 2005.46$ Hz, beating at 10.91 Hz, pass through the fast-element low-pass ($\tau = 0.2$ ms) into the recurrent column. *(a)* The input amplitude spectrum has lines only at f_1 and f_2; in the 8–14 Hz band it is 2.2×10^{-13}, zero to numerical precision. *(b)* The output carries a demodulated line at the beat, 10.9 Hz.

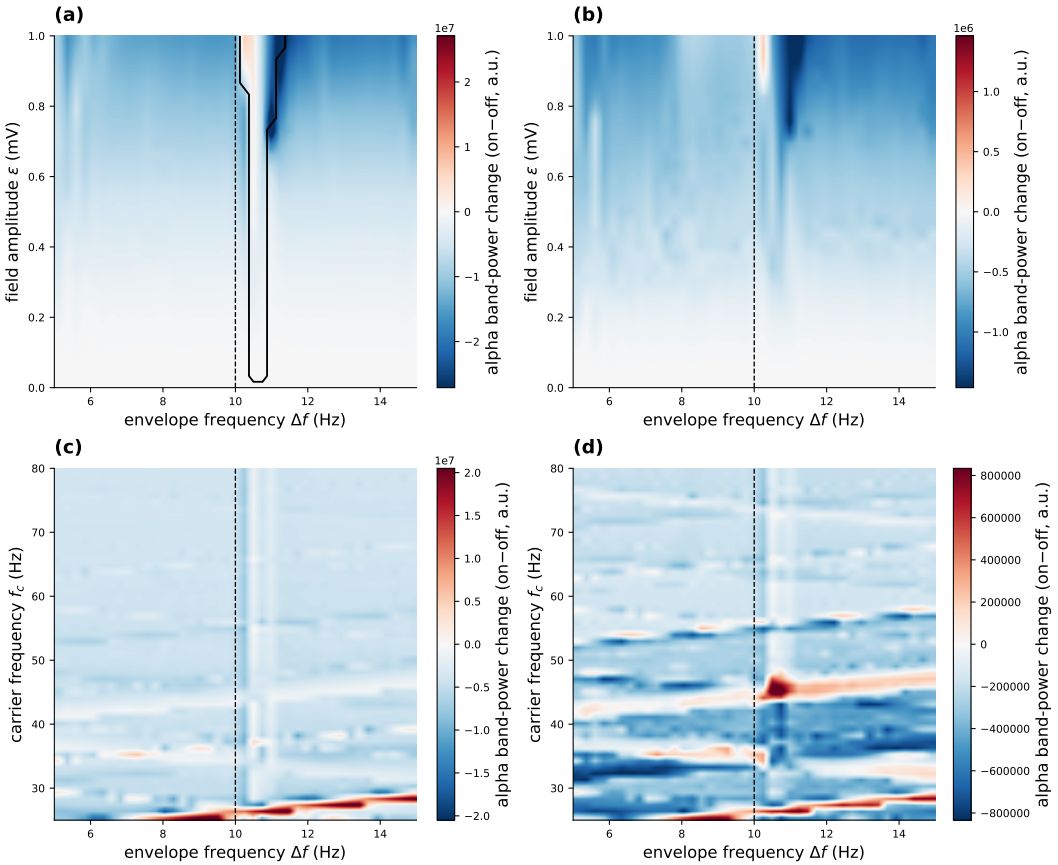


Figure S7: LaNMM Arnold tongues: two-tone TI field delivered directly to the deep P_1 (alpha) loop. As Fig. S1 but with the field on P_1. The deep loop is captured by a much weaker field than the superficial one, so the amplitude range is $\varepsilon \leq 1$ mV here against 3 mV there; above ≈2 mV the drive captures essentially the whole Δf sweep and leaves the perturbative regime the theory describes. Color: signed change of alpha-band power from the $\varepsilon = 0$ baseline, zero-centered; most of the grid is suppression. *(a,b)* P_1 and P_2 enhancement vs. ε and Δf; the black contour in *(a)* outlines the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix F), computed from the dominant output frequency because band power alone cannot separate a driven from an autonomous rhythm; the $\varepsilon = 0$ row is excluded (resolution floor 0.5 Hz). The set widens to 1.25 Hz at $\varepsilon = 1$ mV. *(c,d)* Enhancement vs. carrier frequency and Δf at $\varepsilon = 0.5$ mV. P_1 responds across a *broad* carrier band—a vertical ridge at $\Delta f \approx 10$ Hz, the direct-coupling limit of Eq. (9). Read with the lock-in at Δf the response is retained at every carrier from 25 to 80 Hz (median 25% of peak), in contrast to the ∼7% collapse off resonance under superficial drive.

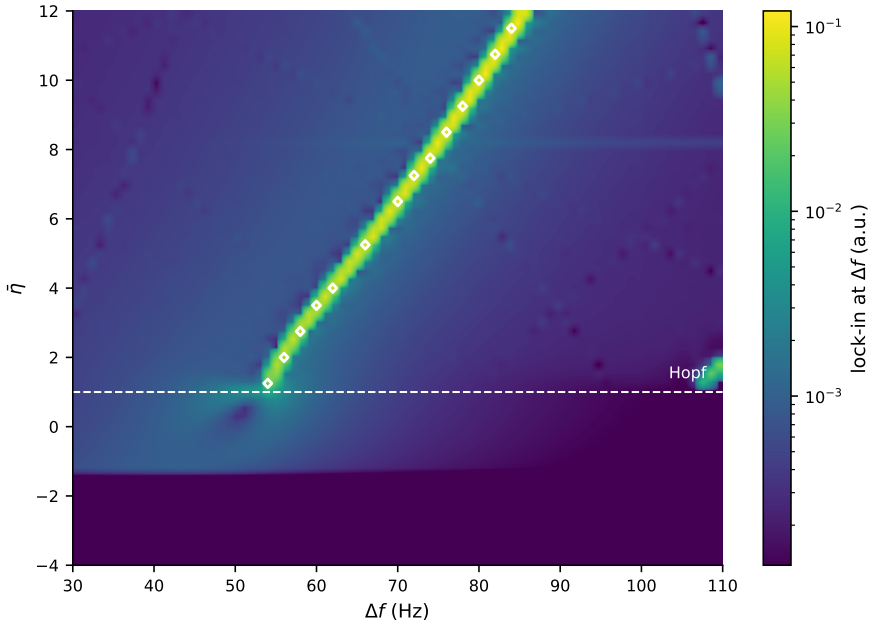


Figure S8: NMM2 PING resonance map (exact mean field). Demodulated response over the (Δf, background drive $\bar{\eta}$) plane, log color scale. The ridge follows the gamma natural frequency, and its diagonal slope—f_0 rising with $\bar{\eta}$—reflects the state-dependent transfer of NMM2, absent in the static-sigmoid models. White markers outline the 1:1 captured set (Appendix F), drawn only above the Hopf (dashed line) where there is an autonomous rhythm to capture; below it the same color field is a forced response. Lock-in amplitude cannot separate the two, since both place power at Δf.

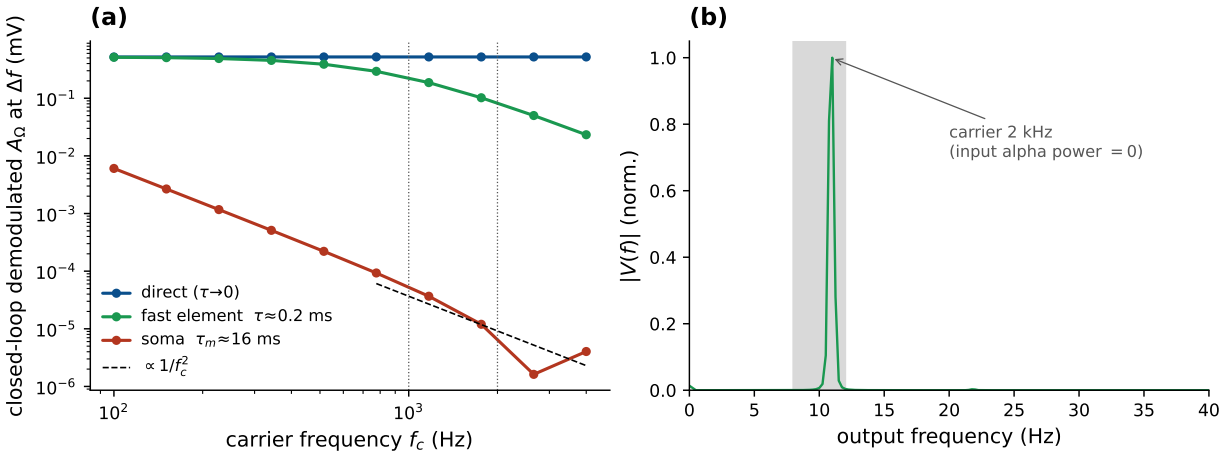


Figure S9: Demodulation at a genuine kHz carrier (Jansen–Rit, closed loop). The applied field passes through an explicit membrane filter before the sigmoid. *(a)* Closed-loop demodulated A_Ω at $\Delta f = f_0$ vs. carrier f_c for three coupling routes: direct ($\tau \to 0$, carrier-independent), fast element (axon/AIS/node, $\tau \approx 0.2$ ms; survives into the kHz band), and soma ($\tau_m \approx 16$ ms; collapses as $1/f_c^2$). *(b)* Output spectrum for a 2 kHz carrier delivered through the fast element: a clean alpha line at f_0, synthesized by the column although the input carries no alpha power.

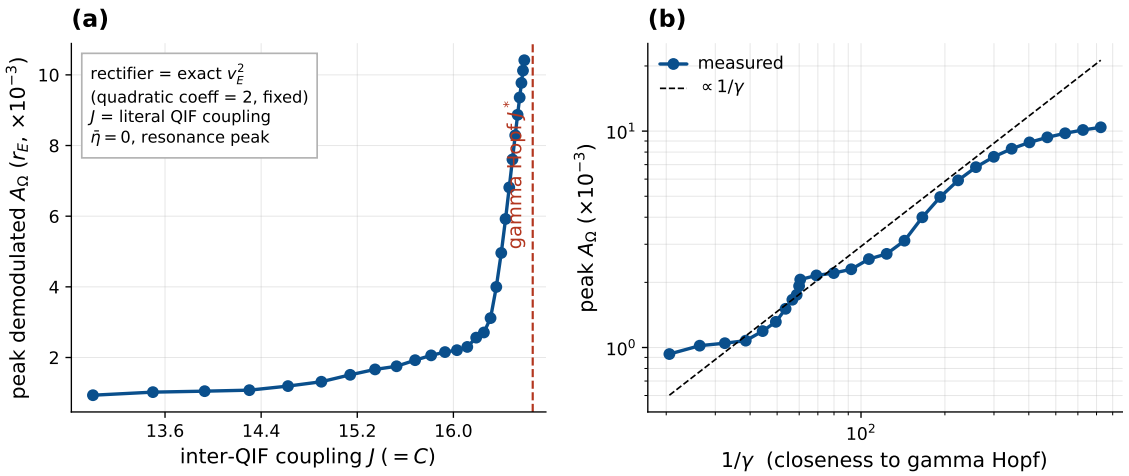


Figure S10: Coupling controls the amplification in the exact mean field (NMM2 PING, gamma). $J\,(=C)$ is the literal inter-QIF synaptic weight; at this carrier the generated source is operating-point independent (§2.3), so no pinning is needed. *(a)* The peak demodulated response of r_E grows as $J \to J^*$ (gamma Hopf, $\bar{\eta} = 0$; stable focus, $\gamma > 0$). *(b)* The gain grows as $1/\gamma$ (dashed), saturating near criticality—the JR law, now with coupling and rectifier both derived.

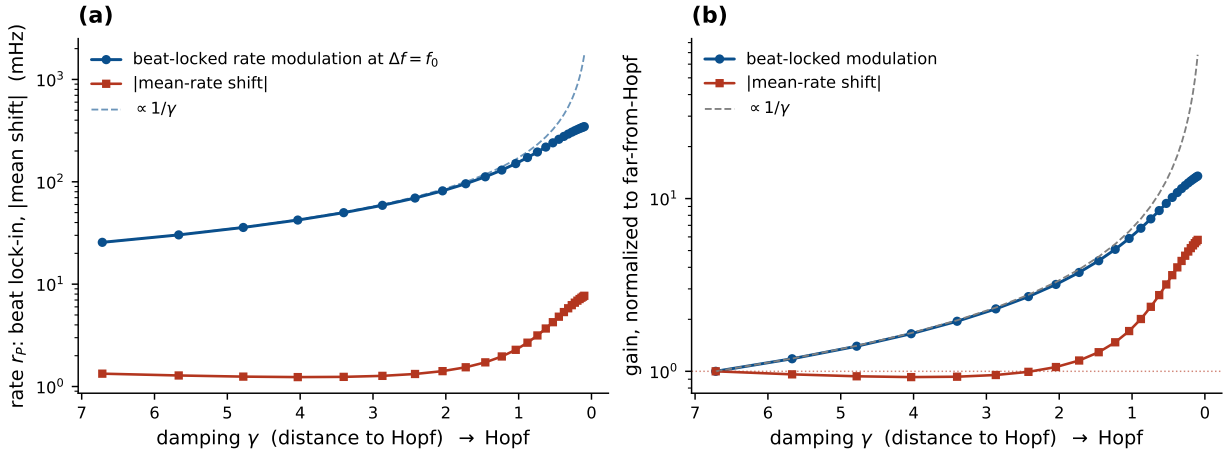


Figure S11: Envelope-locked modulation against mean-rate shift (Jansen–Rit, detection pinned). As the column is driven toward the Hopf (from the stable side, $\gamma > 0$) through the coupling at fixed $\sigma''(v^*)$, both quantities are read from the pyramidal rate $r_P = \sigma(v + s)$, in mHz. *(a)* The beat-locked modulation at $\Delta f = f_0$ (blue) is resonantly amplified and tracks $1/\gamma$ (dashed) before saturating; the magnitude of the mean-rate shift (red) stays at a few mHz and rises near the Hopf. *(b)* Both normalized to their far-from-Hopf values (log scale): the modulation gain follows $1/\gamma$; the mean-shift gain rises later and less. Modulation exceeds mean shift by 19× far from the Hopf and 45–66× near it.

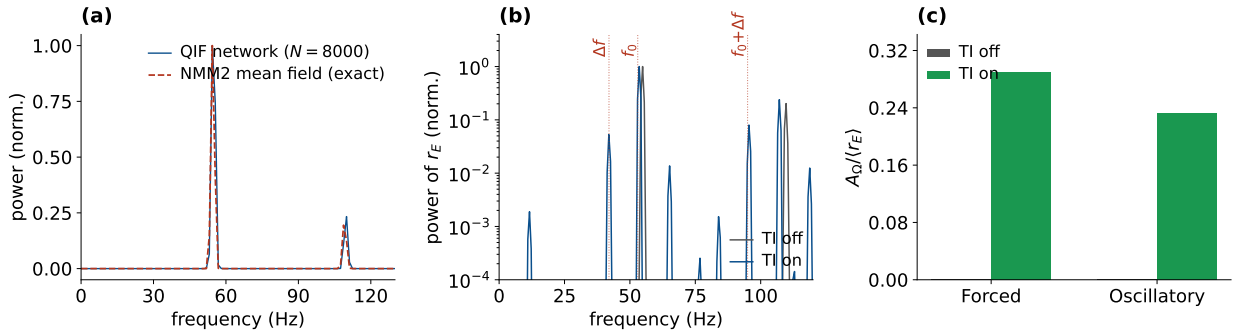


Figure S12: Timing, not rate, in the QIF network. *(a)* Field-free power spectrum of $r_E(t)$, normalized: the spiking network ($N = 8000$) matches the exact NMM2 mean field in gamma frequency (∼53 Hz); a normalized spectrum does not test the mean rate. *(b)* Intermodulation above the Hopf (TI off vs. on, log power): the nonlinearity mixes the imposed $\Delta f = 42$ Hz beat with the intrinsic 53 Hz gamma, raising the direct Δf line and the sum sideband near 95 Hz, both absent with TI off. This is the sideband signature of gating rather than entrainment. *(c)* The Δf modulation depth $A_\Omega/\langle r_E \rangle$, off vs. on: below 0.003 rising to 0.29 (forced) and 0.23 (oscillatory), while the mean rate rises by 8–10%. Modulation depth is reported rather than the off→on lock-in ratio, whose denominator is the field-free floor.

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